The Characteristic Equation

Geometric Algorithms

CAS CS 132

Objectives

- Briefly recap eigenvalues and eigenvectors
- Get a primer on **determinants**
- Determine how to **find eigenvalues** (not just verify them)

Keyword

eigenvectors

eigenvalues

eigenspaces

eigenbases

determinant

characteristic equation

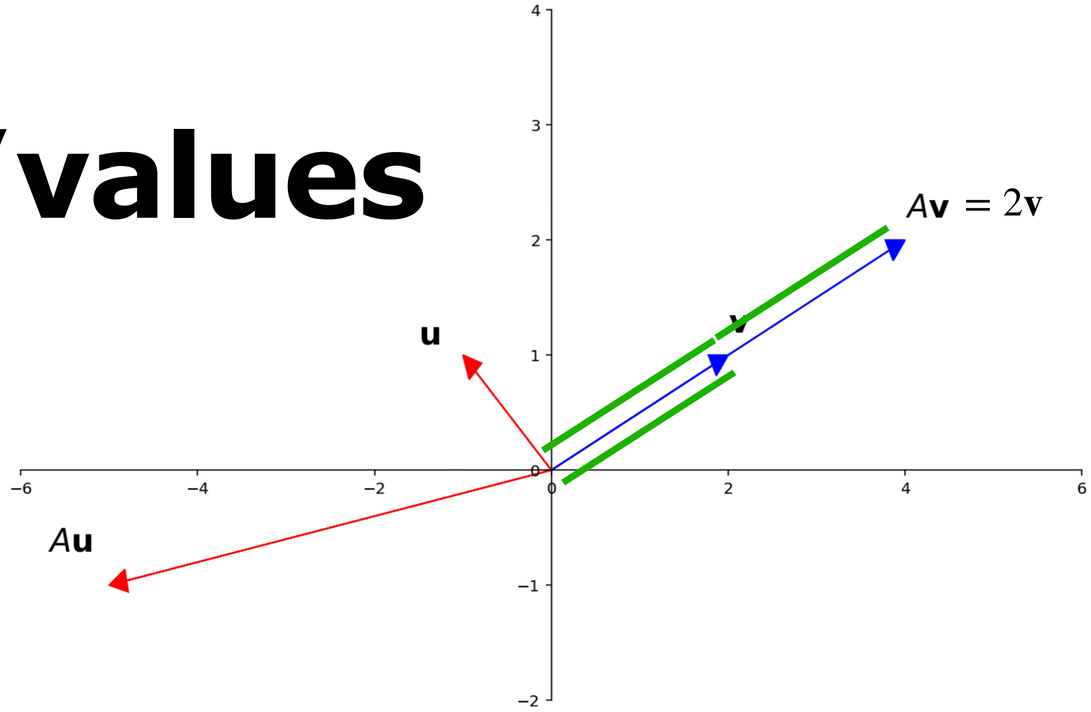
polynomial roots

triangular matrices

multiplicity

Recap

Recall: Eigenvectors/values



Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector and eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if

$$A\mathbf{v} = \lambda \mathbf{v}$$

Eigenvectors are just scaled by A

Recall: Verifying Eigenvectors

Q. Determine if $\mathbf{v}$ is an eigenvector of A and determine the corresponding eigenvalues

A. Easy. Work out the matrix-vector multiplication

Ex.

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 13 \\ 9 \end{bmatrix}$$



Recall: Verifying Eigenvalues

Q. Find an eigenvector of A whose eigenvalue is λ

A. Find a nontrivial solution to

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

If we don't need the vector we can just show that $A - \lambda I$ *is* ***not*** *invertible (by IMT)*

Recall: Finding Eigenspaces

Q. Find a basis for the eigenspace of A for λ

A. Find a basis for $\text{Nul}(A - \lambda I)$

(we did this for our recap problem)

Practice Problem

$$\begin{bmatrix} 5 & 2 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 2 & 4 & 10 & 0 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

Determine the dimension of the eigenspace of A *for the eigenvalue 4*

(try not to do any row reductions)

Answer

$$\begin{bmatrix} 5 & 2 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 2 & 4 & 10 & 0 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

Recall: Eigenvectors and Linear Dynamical Systems

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if $\mathbf{v}_0$ can be written in terms of eigenvectors $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k$ of A with eigenvalues $\lambda_1 > \lambda_2 \ldots \geq \lambda_k$ then $\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$ for some constant c_1

Eigenvalues of Triangular Matrices

Thm. The eigenvalues of a triangular matrix are its entries along the diagonal

$$\begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

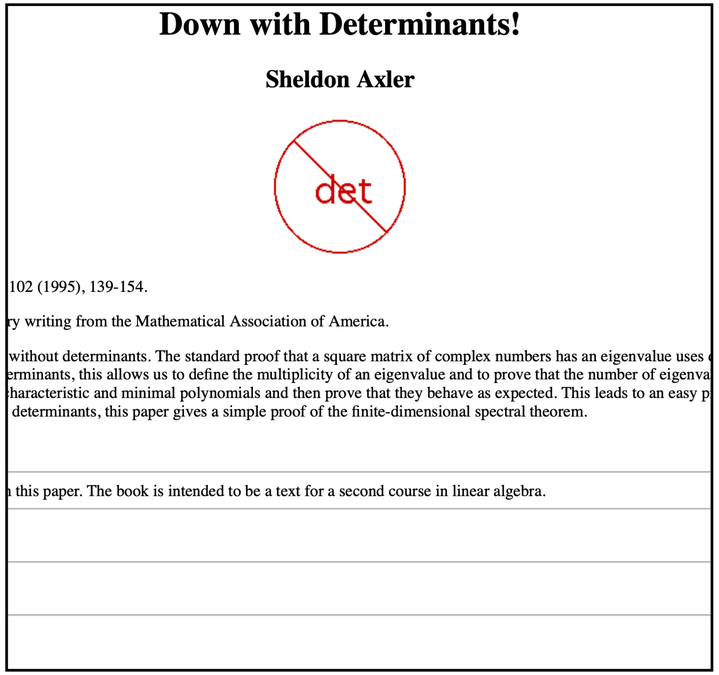
Determinants

An Aside: Determinants are Mysterious

Determinants are strangely polarizing

Some people love them, some people hate them

We'll only scratch the surface...



Down with Determinants!

Sheldon Axler

102 (1995), 139-154.

ry writing from the Mathematical Association of America.

without determinants. The standard proof that a square matrix of complex numbers has an eigenvalue uses
erminants, this allows us to define the multiplicity of an eigenvalue and to prove that the number of eigenva
haracteristic and minimal polynomials and then prove that they behave as expected. This leads to an easy p
determinants, this paper gives a simple proof of the finite-dimensional spectral theorem.

n this paper. The book is intended to be a text for a second course in linear algebra.

What kind of thing is the determinant?

A determinant, written $\det(A)$, is a *number* associated with a matrix

In broad strokes, it's a big sum of products of entries of A

A Scary-Looking Definition (we won't use)

$$\det(A) = \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)}$$

We can think of this as a procedure

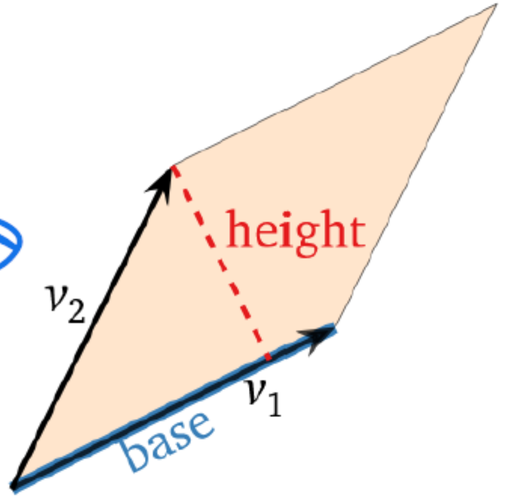
```
1 FUNCTION det(A):
2   total = 0
3   FOR all matrix B we can get by swapping a bunch of rows of A:
4     s = 1 IF (# of swaps necessary) is even ELSE -1
5     total += s * (product of the diagonal entries of B)
6   RETURN total
```

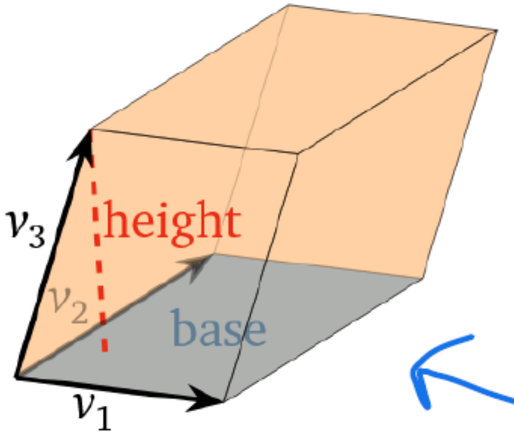
A Geometric Interpretation: Volume

$$|\det[\vec{v}_1\ \vec{v}_2]|$$

$$\|$$

$$\text{vol}(P)$$





$$|\det[\vec{v}_1\ \vec{v}_2\ \vec{v}_3]|$$

$$\|$$

$$\text{vol}(P)$$

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$(-1)^0 ad$ is the product of the diagonal

$(-1)^1 bc$ is the product of the diagonal after 1 swap

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow^{0} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$(-1)^0 ad$$

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow^{1} \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

$$(-1)^{1} cb$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

The Determinant of 3×3 matrices

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$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{0} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$(-1)^{0} aei$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{2} \begin{bmatrix} g & h & i \\ a & b & c \\ d & e & f \end{bmatrix}$$

$$(-1)^2 gbf$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{2} \begin{bmatrix} d & e & f \\ g & h & i \\ a & b & c \end{bmatrix}$$

$$(-1)^2 dhc$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{1} \begin{bmatrix} g & h & i \\ d & e & f \\ a & b & c \end{bmatrix}$$

$$(-1)^{1} gec$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{1} \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix}$$

$$(-1)^{1} dbi$$

The Determinant of 3×3 matrices

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^{1} \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}$$

$$(-1)^{1} ahf$$

Another Perspective

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{R_2 \leftarrow aR_2} \begin{bmatrix} a & b \\ ac & ad \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - cR_1}$$

$$\sim \begin{bmatrix} a & b \\ 0 & ad-bc \end{bmatrix}$$

$ad - bc = 0 \iff A$ is invertible

Another Perspective

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \sim \begin{bmatrix} a & b & c \\ 0 & ae-db & af-dc \\ 0 & ah-gb & ai-gc \end{bmatrix}$$

B

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\begin{bmatrix} a & b & c \\ 0 & ae-db & af-dc \\ 0 & 0 & \det(B) \end{bmatrix}$$

ex.

$$\det(B) = \det(A)$$

Determinants and Invertibility

Thm. A is invertible if and only if $\det(A) \neq 0$

So we can extend the IMT:

- A is invertible
- $\det(A) \neq 0$
- 0 is not an eigenvalue

(these must be all true or all false)

Determinants (the definition we'll use)

$$\det(A) = \frac{(-1)^s}{c} U_{11}U_{22}\ldots U_{nn}$$

product of diagonal entries

0 if A is not invertible

Df. The **determinant** of a matrix A is given by the above equation, where

» U is an *echelon form* of A

» s is the number of row *swaps* used to get U

» c is the *product of all scalings* used to get U

Example

$$\begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

The definition holds no matter which row operations you use

How To: Determinants

Q. Determine the determinant of a matrix A

A.

1. Convert A to an echelon form U
2. Keep track of the number of row swaps you used, call this s, and the product of all scalings, call this c
3. Determine the product of entries along the diagonal of U, call this P
4. The determinant of A is $\frac{(-1)^s P}{c}$

Another way: cofactor expansion

(Useful when certain rows/columns are sparse)

Expansion along row i:

$$\det(B) = \sum_{j=1}^{n} (-1)^{i+j} b_{i,j} m_{i,j},$$

Expansion along column j:

$$\det(B) = \sum_{i=1}^{n} (-1)^{i+j} b_{i,j} m_{i,j}.$$

(notation stolen from "Laplace expansion" Wikipedia page)

Example (Again)

$$\begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

The Shorter Version

numpy.linalg.det(A)

Beyond small matrices, we'll just use a computer

Properties of Determinants

Properties of Determinants (1)

$$\det(AB) = \det(A)\det(B)$$

Cor. AB is invertible $\Leftrightarrow$ A and B are invertible

Question

Use the fact that $\det(AB) = \det(A)\det(B)$ *to give an expression for* $\det(A^{-1})$ *in terms of* $\det(A)$

Hint. What is $\det(I)$*?*

Properties of Determinants (2)

$$\det(A^T) = \det(A)$$

It follows that A^T is invertible if and only if A is invertible

(we also won't verify this)

Question

If $A^{-1} = A^T$*, then what is* $\det(A)$*?*

Properties of Determinants (3)

Thm. If A is triangular, then $\det(A)$ is the product of entries along the diagonal

Question

$$\begin{bmatrix} 1 & 5 & -4 \\ -1 & -5 & 5 \\ -2 & -8 & 7 \end{bmatrix}$$

Find the determinant of the above matrix

Answer

$$\begin{bmatrix} 1 & 5 & -4 \\ -1 & -5 & 5 \\ -2 & -8 & 7 \end{bmatrix}$$

Characteristic Equation

Finding Eigenvalues

Q. Determine the eigenvalues of A, along with their associated eigenspaces

A. Can we somehow "solve for λ" in the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

What kind of thing is the determinant, really?

The determinant of a matrix A is an *arithmetic expression* written in terms of the entries of A

But a matrix may not have numbers as entries

We might think of the matrix $A - \lambda I$ has having *polynomials* as entries

Then $\det(A - \lambda I)$ is a **polynomial**

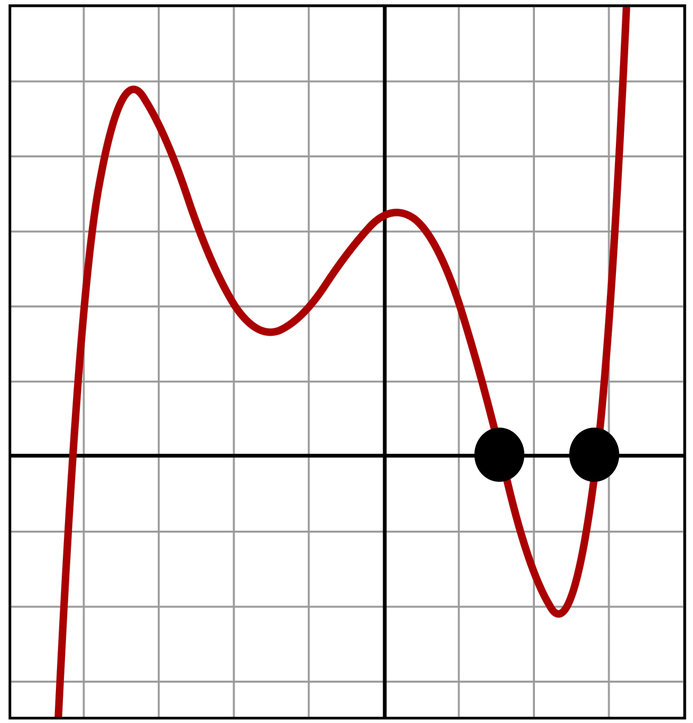
Reminder: Polynomial Roots

A **root** of a polynomial $p(x)$ is a value r such that $p(r) = 0$

(A polynomial may have many roots)

If r is a root of $p(x)$, then it is possible to find a polynomial $q(x)$ such that

$$p(x) = (x - r)q(x)$$



https://commons.wikimedia.org/wiki/File:Quintic_polynomial.svg

Characteristic Polynomial

Df. The **characteristic polynomial** of a matrix A is $\det(A - \lambda I)$ viewed as a polynomial in the *variable* λ.

This is a polynomial with the eigenvalues of A as roots

So we can "solve" for the eigenvalues in the equation

$$\det(A - \lambda I) = 0$$

Example: 2×2 **Matrix**

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

A Special Linear Dynamical System

$$\mathbf{v}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_k \qquad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

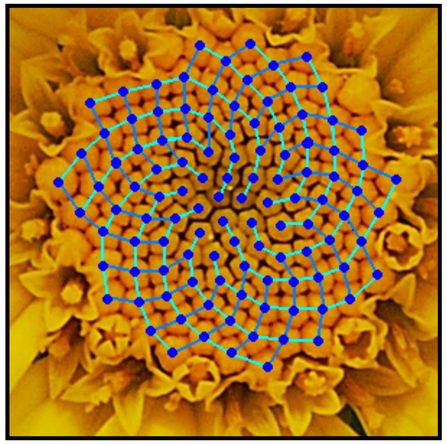
Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_k = F_{k-1} + F_{k-2}$$

```
define fib(n):
  curr, next ← 0, 1
  repeat n times:
    curr, next ← next, curr + next
  return curr
```

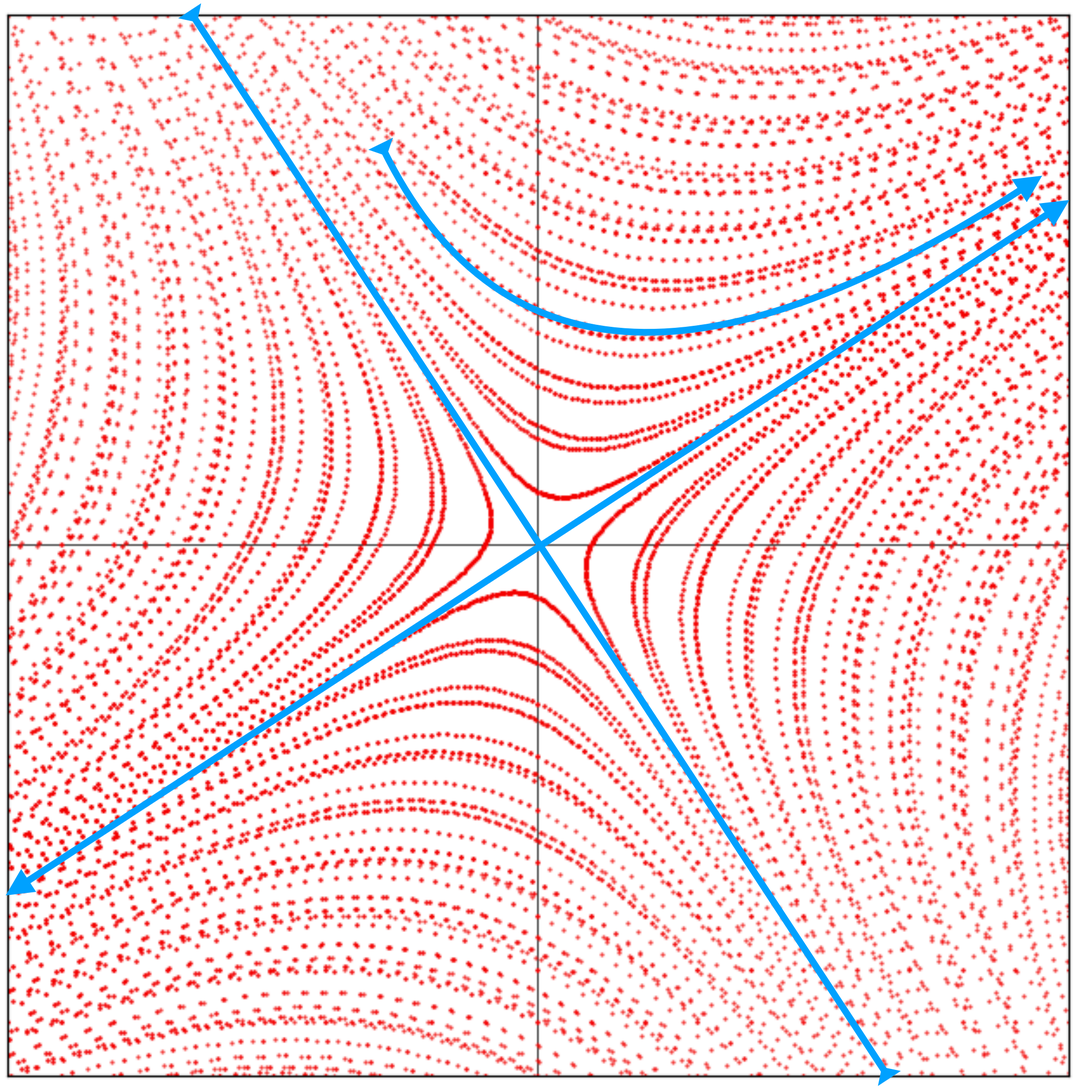


The Fibonacci numbers are defined in terms of a recurrence relation

They seem to crop-up in nature, engineering, etc.

https://commons.wikimedia.org/wiki/File:FibonacciChamomile.PNG

Recall: The Picture



The eigenvalue of this matrix is the golden ratio

Golden Ratio

$$\varphi = \frac{1+\sqrt{5}}{2} \qquad \frac{F_{k+1}}{F_k} \to \varphi \text{ as } k \to \infty \qquad F_k \sim c\varphi^k$$

This is the largest eigenvalue of $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

The "long term behavior" is the Fibonacci sequence is defined by the golden ratio

Example: Triangular matrix

$$\begin{bmatrix} 1 & -3 & 0 & 6 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

How To: Finding Eigenvalues

Q. Find all eigenvalues of the matrix A

A. Find the roots of the characteristic polynomial of A

An Observation: Multiplicity

$$\lambda^1(\lambda-1)^2(\lambda-4)^1$$ multiplicities

In the examples so far, we've seen a number appear as a root multiple times

This is called the **algebraic multiplicity** of the root

Is the algebraic multiplicity meaningful?

Multiplicity and Dimension

Thm. The dimension of the eigenspace of A for the eigenvalue λ is *at most* the algebraic multiplicity of λ in $\det(A - \lambda I)$

The algebraic multiplicity is an upper bound on "how large" the eigenspace is

Example

Let $A \in \mathbb{R}^{5\times5}$ with characteristic polynomial

$$(x-1)^3(x-3)(x+5)$$

» What is $\text{rank}(A)$?

» What is the minimum possible rank of $A - I$?